MIL-D-22955(SHIPS)
15 September 1961

MILITARY SPECIFICATION

DOMES, SONAR, STEEL, TYPES CW-553/SQS, CW-554/SQS AND CW-559/SQS AND MOUNTINGS, SONAR DOME, MT-2327/SQS, MT-2328/SQS AND MT-2369/SQS

1. SCOPE

1.1 Scope. - This specification covers the construction of steel sonar domes and sonar dome mountings for installation aboard Naval ships.

1.2 Classification. - Steel sonar domes and sonar dome mountings shall be furnished as specified (see 6.2), in the following types:

Dome	Mounting
CW-553/SQS	MT-2327/SQS
CW-554/SQS	MT-2328/SQS
CW-559/SQS	MT-2369/SQS

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel, Plates, Sheets, and Strip--Corrosion Resisting.

FSC 5845

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MIL-Q-9858 - Quality Control System Requirements
MIL-P-15024 - Plates, Identification--Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-P-15929 - Primer, Paint, Vinyl-Red Lead Type Formula No. 119.
MIL-P-15930 - Primer, Paint, Vinyl Zinc Chromate Type, Formula No. 120.
MIL-P-15931 - Paint, Anti-Fouling, Vinyl-Red (Formula No. 121).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-A-18001 - Anodes, Corrosion Preventive, Zinc: Plate, Slab, Disc and Rod Shaped.
MIL-E-21981 - Electronic Type Designations, Identification Plates and Markings, Requirements for
MIL-N-25027 - Nut, Self-Locking, 250°F., 500°F., and 800°F.

STANDARDS

MILITARY

MIL-STD-10 - Surface Roughness Waviness and Lay.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS

RE78J258 - Dome, Sonar, Steel, 100-inch, Type CW-553/SQS.
RE78J259 - Dome, Sonar, Steel 185-inch, Type CW-554/SQS.
Microfilm Reel No. 26935 - Dome, Sonar, Steel, 360-inch, Type CW-559/SQS.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
A7-58T - Steel for Bridges and Buildings.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York)

3. REQUIREMENTS

3.1 Materials. - Unless otherwise specified herein, materials used in the construction of sonar domes CW-553/SQS, CW-554/SQS and CW-559/SQS and mountings MT-2327/SQS, MT-2328/SQS and MT-2369/SQS shall be as shown on Drawings RE78J258, RE78J259, and Microfilm Reel No. 26935, respectively.

3.1.1 Corrosion-resisting steel. - Corrosion-resisting steel shall conform to class 304 of Specification QQ-S-766. Corrosion-resisting steel used in the skin of all sonar domes shall have a number 1 finish on both sides, except that such steel shall have a number 2-D finish on both sides where the thickness is less than 0.050 inch.

3.1.2 Hot-rolled steel. - Hot-rolled steel shall conform to ASTM Standard A7-58T with a maximum carbon content of 0.35 percent. The use of bessemer and resulfurized steel is not acceptable.

3.1.3 Anodes. - Zinc anodes shall conform to class 1 of Specification MIL-A-18001

3.2 Deviations from drawing. - The following requirements apply only to Microfilm Reel No. 26935.

(a) The flat steel washer (piece 27), and the Fabreeka washer (piece 31) shall be circular with an outside diameter of 3-1/2-inches and a central hole of 1-1/64 inches diameter. Piece 27 shall be 3/8 inch thick.

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(b) Delete the lockwasher (piece 17).
(c) Delete the rubber gasket (piece 30).
(d) The corrosion-resisting skin shall extend a maximum of 1/4 inch above the lower edge of the bottom plate (piece 2-2), and shall be welded as shown on the drawing. At no point on the periphery of the dome shall the distance between the toe of this weld and the upper edge of piece 2-2 be less than 1/4 inch.
(e) Delete the 1/4-inch by 1/2-inch round head machine screws (piece 36).
(f) The four grounding straps (piece 28) shall be manufactured of 18 gage corrosion-resisting steel, and shall be 2 inches wide by 7 inches long without holes in either end. One end of each grounding strap shall be shaped to fit and welded to the edge of individual flat washers (piece 27). When assembling, these flat washers, with grounding straps attached, shall be installed approximately equally spaced around the dome. The grounding straps shall then be bent into a "U" shape and the free ends welded to the nearest gusset (piece 2-8). Care shall be taken to prevent the grounding strap from touching any surface in a manner which might result in rattling. The grounding strap shall not pass between the two mounting bolts located between gussets.
(g) A fairing strip shall be manufactured but not installed by the contractor. This fairing strip shall be of hot-rolled steel, 11-gage, 5 inches wide by 786 inches long. The fairing strip may be fabricated from shorter lengths provided the butt welds are ground flush on both sides and edges. The fairing strip shall be given one coat of pretreatment, formula 117 (see 3.12.1).
(h) The 1/2-inch by 4-1/2-inch by 62-feet, 1-inch bar (piece 2-3) shall be located flush with the inner edge of the upper plate (piece 2-1), beveled 3/8-inch by 60 degrees on the upper, inner edge and welded in a manner similar to that shown between the vertical side and flange of the sonar dome mounting. The gusset (piece 2-8) shall be increased in size sufficient to allow welding as shown.
(i) A single layer of 9/16-inch thick B. F. Goodrich SAPER D-2 rubber, or equivalent, shall be used in lieu of single and double thicknesses of SAPER rubber in both the after curved baffle and the overhead flat baffle.

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3.2.1 Self-locking nuts. - For the CW-553/SQS and CW-554/SQS domes only, all metal, corrosion-resisting steel, self-locking hexagon nuts shall be used in lieu of pieces 21 and 22 of Drawings RE78J258 and RE78J259, respectively. These nuts shall conform to Specification MIL-N-25027 and Specification MIL-E-16400.

3.3 General requirements. - Unless otherwise specified herein, the design, contours, dimensions, and requirements for the Sonar Domes CW-553/SQS, CW-554/SQS and CW-559/SQS and mountings MT-2327/SQS, MT-2328/SQS and MT-2369/SQS shall be as shown on Drawings RE78J258, RE78J259 and Microfilm Reel No. 26935, respectively.

3.4 Dome and mounting flanges. - The surfaces of the dome flange and the mating mounting flange shall be within plus or minus 1/32 inch of their planes.

3.5 Adapter flange. - The bolting surfaces of the adapter flange for the Sonar Dome CW-554/SQS shall be flat with plus or minus 1/32 inch of their planes and shall be parallel to within plus or minus 1/64 inch.

3.6 Interchangeability - All domes and mountings of the same type produced under the contract or order shall be interchangeable with each other and with existing domes and mountings of the same type (see 3.6.1).

3.6.1 Bolting circle template. - A template of the bolting circles of the dome and mounting flanges for each type shall be used by the contractor. This template will be supplied by the Government.

3.7 Transducer mounting flange. - The transducer mounting flange within the dome to which either the transducer or the adapter flange mounts (see 3.4), shall be within plus or minus 1/32 inch of its mean plane. The mean plane of this flange shall be parallel within plus or minus 1/32 inch of the mean plane of the dome flange (see 3.4).

3.8 Dome skin contours and smoothness. - The outside surface of the mounting, and the skin and the welded seams of the dome shall be kept as close to the required streamlined contours as is economically feasible under production techniques. In no case shall the tolerances vary by more than plus or minus 1/16 inch over any 6 inch square area of the prevailing contour, nor shall they vary by more than plus or minus 1/64 inch in any 2 inch square along the forward half of the dome. There shall be no sharp edges or irregularities over the exterior surface such as may be caused by welds and seams. Any heat treatment of the skin to meet acceptable tolerances will not be permitted.

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3.9 Welding. - Welding shall conform to Standard MIL-STD-278.

3.10 Degreasing. - All grease or oil on the dome and mounting shall be removed by using an emulsifying type of cleaner followed by a water rinse at a temperature of no less than 130°F.

3.11 Surface finish. - A pressure wet blast with abrasive slurry shall be used on the corrosion-resisting steel portions of the dome to obtain a roughness finish of 125 to 250 as specified in Standard MIL-STD-10.

NOTE: Pressure wet blast, using a slurry of abrasive and water, is preferred in order to obtain the necessary clean surface. Dry blasting results in embedding particles of the abrasive into the corrosion-resisting steel. These particles cannot be seen with the naked eye but are quite visible at 12X to 36X magnification. These particles of abrasive may induce early failure of the paint film through blistering and loss of adhesion, and may also promote corrosion by inducing the corrosion-resisting steel to become active in relation to surrounding areas. After the corrosion-resisting steel dome has been blasted, the surface shall not be touched and foreign matter shall no be allowed to come in contact with the surface. All exterior surfaces shall be covered (clean paper will be suitable) if the dome is to be shipped to another location. Masking tape shall not be used directly on any of the blasted surfaces. When the dome is ready for painting the covering shall be carefully removed and the painting started immediately. All painting shall be accomplished under shop conditions with a clean atmosphere and a temperature of at least 60°F.

3.12 Paint. - The painting instructions of this section apply to all interior surfaces of the dome and mounting, the surfaces of the dome flange and mating mounting flange, and the mild steel portion of the mounting and dome exterior. The corrosion-resisting steel portion of the dome exterior shall not be painted by the manufacturer.

3.12.1 Spray apply 1 coat of pretreatment coating, formula 117 as specified in Specification MIL-C-15328. This coat shall be applied in two even, crossed passes of sufficient thickness to obtain a film thickness when dry of 0.7 mil. Formula 117 shall appear "wet", not dry or powdery, as it is applied. Prior to mixing the formula 117, all of the settled pigment in the resin component shall be redispersed. Actual examination of the emptied 5-gallon container after mixing is suggested to be certain there is no pigment left. All of the zinc chromate pigment is required for proper paint adhesion. The mixed pretreatment shall be allowed to stand for 1/2 hour, and the pigment then shall be redispersed before using.

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3.12.2 The surface shall be dried under mild heat, such as lamps, at a surface temperature of 100° to 110°F.

3.12.3 The second coat shall be vinyl zinc chromate paint primer, formula 120, in accordance with Specification MIL-P-15930. A wet coat shall be applied of sufficient thickness to obtain a film thickness when dry of 1.5 mils.

3.12.4 Drying as specified in 3.12.2 shall be repeated.

3.12.5 The third coat shall be vinyl red lead paint primer, formula 119, in accordance with Specification MIL-P-15929. A wet coat shall be applied of sufficient thickness to obtain a film thickness when dry of 1.5 mils.

3.12.6 Drying as specified in 3.12.2 shall be repeated.

3.12.7 The fourth coat shall be formula 120 in accordance with Specification MIL-P-15930 applied and dried as specified in 3.12.3 and 3.12.2.

3.12.8 The fifth coat shall be formula 119 in accordance with Specification MIL-P-15929 applied and dried as specified in 3.12.5 and 3.12.2.

3.12.9 The sixth coat shall be vinyl-red anti-fouling paint, formula 121, in accordance with Specification MIL-P-15931, or the improved anti-fouling paint, formula 121X. A wet coat shall be applied of sufficient thickness to obtain a dry film thickness of 2 mils.

NOTE: Formula 121X is a higher solids paint than standard 121, requiring less material to obtain 2 mils dry film thickness. Formula 121X shall be used as soon as available, since it is preferred to formula 121. Local procurement may be made in accordance with the following formula, which is a modification of the formula given in Specification MIL-P-15931:

Ingredient	Pounds
Cuprous oxide	1600
Rosin	240
Vinyl resin	60
Tricresyl phosphate	56
Methyl isobutyl ketone	235
Xylene	158
Viscosity should be between 75-85 Krebs unit	

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3.12.10 Drying as specified in 3.12.2 shall be repeated.

3.12.11 The seventh and final coat shall be formula 121 or 121X, applied as specified in 3.12.9.

3.12.12 The final coat shall be allowed to air dry for a minimum of 4 hours before handling.

3.12.13 The complete anti-fouling paint system shall have a dry film thickness of 10.7 mils. The following are the thicknesses of the various paints in the system:

1 coat 117 ----------	0.7 mils.
2 coats 119 ----------	3.0 mils.
2 coats 120 ----------	3.0 mils.
2 coats 121 or 121X --------	4.0 mils.
Total	10.7 mils.

3.12.14 The zinc anodes shall not be painted. Zincs shall be left clean and bright.

3.13 Indentification plates. - Corrosion-resisting steel identification plates bearing markings and assigned designations shall be furnished in accordance with type type A, B, or C of Specification MIL-P-15024, except that the attachment of the plate shall be as specified hereinafter. The identification plate for each dome shall be attached to the corrosion-resisting steel skin just below the mounting at the stern end. Immediately above this, on the mounting, the identification plate for each sonar dome mounting shall be attached. Attachment may be accomplished by means of a weld or by soldering sufficient to hold the plate to the skin under all anticipated sea conditions.

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3.14 Deviations.- Minor deviations from Drawings RE78J258, RE78J259 and Microfilm Reel No. 26935 necessitated by shop practices, and deemed essential to ensure superior workmanship, will be permitted, subject to the approval of the bureau or agency concerned as having no adverse effects upon the required material performance. Any such minor changes which are approved by the bureau or agency concerned shall be incorporated in the manufacturing drawings.

3.15 Government nomenclature.- Nomenclature shall be requested in accordance with the procedures outlined in Specification MIL-E-21981.

3.16 Workmanship.- Workmanship shall be of high quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Control of quality.- The contractor shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification. The system of quality control shall be in accordance with Specification MIL-Q-9858.

4.3 Welds.- An examination of all welds (see 3.8) used for attaching the corrosion-resisting steel skin shall be made and proved air tight and free from cracks. This examination shall be made by the use of internal air pressure at 3 pounds per square inch gauge, with leaks being detected by the application of a soap solution to the exterior surfaces of the dome. All cracks shall be repaired using a welding process. Examination shall be made prior to painting the domes.

4.3.1 All welds not specified in 4.3 shall be examined on sample domes and mountings selected in accordance with 4.4, and proved free from cracks.

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4.4 Sampling.- Domes and mountings shall be selected in accordance with table I and examined in accordance with the paragraphs indicated:

Table I - Sampling for examination of sonar domes and mountings.

Lot size, number of sonar domes or mountings	Sample size, number of domes or mountings to be examined	
	4.3.1, 4.5, 4.6, 4.7, 4.8	4.9, 4.10
2 to 8	5	2
9 to 15	7	3
16 to 25	10	5
26 to 40	13	6
41 to 65	16	8
66 to 110	21	10
111 to 180	27	12

4.5 Surface irregularities.- In order to assure that no irregularities, sharp edges, or other detrimental contours exist, a thorough visual examination of the acoustic window smoothness shall be made with particular attention to the spot and seam welds throughout the dome. Based on this visual examination, suspected areas shall be checked for conformance with the tolerances specified in 3.8.

4.6 Contour deviations.- Contour deviation shall be determined as follows: Use a flexible steel tape measure (or a suitable substitute) and a depth gauge to check an area suspected of exceeding the allowable variation from the prevailing streamlined contour of the specific dome. Stretch the tape measure over high spots and measure the depth of low spots with a depth gauge. The lower edge of the depth gauge shoulder and the stretched tape measure should be lined up on the suspected 6-inch square. Measurements shall not exceed the allowable tolerance (for example, for a plus or minus 1/16-inch allowable variation from the prevailing contour, the maximum measurement shall not exceed 1/8-inch). The prime areas to inspect for variations are in the vicinity of the welded seams.

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4.7 Skin irregularities. - A depth gauge that has a shoulder approximately 2 inches long shall be used to check tolerances within a 2-inch square over the forward half of the dome skin. With the shoulder of the gauge resting on the high spots of the suspected area, the depth of the low spot shall be measured. Measurement shall not exceed 1/32-inch.

4.8 Sharp projections. - All places on the dome skin suspected of having sharp projections or depressions shall be checked by running a piece of finished steel 3/16-inch, or less, wide and approximately 20 mils thick at the base, across and in contact with the acoustic window at an angle of approximately 45 degrees. The tool shall be run across the suspected area from all directions in the manner of a chisel. Catching of the tool on a surface protrusion or depression indicates an unacceptable sharp edge.

4.9 Weights. - Weights of uncrated and crated domes and mountings shall be determined as specified for weight and dimensions in Specification MIL-E-16400, and forwarded to the contracting officer.

4.10 Determining interchangeability. - Notwithstanding the use of a Government-furnished bolting circle template (see 3.6.1), random measurements or actual interchange of mountings and domes of the same type shall be made of 5 percent of the units in the contract or order, to insure compliance with the interchangeability requirement of 3.6.

4.11 Flatness. - All domes and mounting shall be examined to assure compliance with 3.4 and 3.7.

4.12 The examination called for in 4.3.1, 4.4, 4.5, 4.6, and 4.7 shall be made after the air test and any subsequent repair (see 4.3).

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Preservation and packaging. - Preservation and and packaging of sonar domes and mountings shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

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5.1.2 Packing. - Packing of sonar domes and mountings shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers, or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier Regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment marking information shall be provided on shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal Stock number or manufacturer's part number, type, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Preservation, packaging, packing and marking. - Domes shall be preserved and packaged level A or C; packed level A or B and marked as specified (see 6.2) in accordance with Specification MIL-E-17555.)

6. NOTES

6.1 Intended use. - The equipment covered by this specification is intended for Naval service where it is expected to withstand continuous use for long periods, under Military service conditions, without benefit of overhaul. The equipment is vital. Failure at a critical moment invariably results in serious reduction in the battle efficiency of the ship to which it is attached. Inspection can be performed by a diver, but repair or replacement requires costly and time-consuming drydocking of the ship.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of sonar dome (see 1.2).
(c) Type of mounting (see 1.2).
(d) Preservation, packaging, packing, and marking requirements if other than those specified in 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5845-N186Sh)

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.) Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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